

Schrödinger's equation

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1 Introduction

This short paper will aim to "derive" Schrödinger's equation and give the reader an understanding for its significance and uses. In Prerequisites, I will the fundamentals of the wave theory required to follow the derivation that is detailed in the following part.

2 Prerequisites

2.1 Maclaurin series

First assume that a sufficiently differentiable or "well behaved" function can be written as an infinite power series (a polynomial of infinite degree) such that:

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 \dots$$

now to find the coefficients it should be obvious that:

$$f(0) = a_0$$

and we can use the same method for the other coefficients once differentiating (shown below for the first and second derivative respectively):

$$f'(x) = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 \dots$$

$$f'(0) = a_1$$

$$f''(x) = +2a_2 + 6a_3x + 12a_4x^2 \dots$$

$$f''(0) = 2a_2$$

now it is apparent that we can generalise the rule and make the following deductions

$$f^n(0) = n!a_n$$

$$a_n = \frac{f^n(0)}{(n)!}$$

and thus we can re-write our original function

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots$$

and this is the Maclaurin (or power) Series.

2.2 Euler's formula

applying this formula for e^x the derivative remains the same so

$$f^i(0) = e^0 = 1$$

for all $i \in n$. And thus

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

applying this formula for sine and cosine we get:

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

the derivations of these are left as an exercise to the reader, but you should already see where the opposing signs and parities come from. now If we let $x = i\theta$

$$\begin{aligned} e^{i\theta} &= 1 + (i\theta) + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \dots \\ &= 1 + (i\theta) - \frac{\theta^2}{2!} - \frac{i\theta^3}{3!} + \frac{\theta^4}{4!} + \dots \\ &= \cos(\theta) + i\sin(\theta) \end{aligned}$$

2.3 wave equations

In this section we are going to show and explain why we can represent a wave in the form

$$\psi = Ae^{i(kx - \omega t)}$$

(I will use w instead of ω because it is easier to write) from the previous parts we derived Euler's formula

$$e^{i\theta} = \cos(\theta) + i\sin(\theta)$$

which shows that an exponential is the combination of an imaginary and real wave. And having it in this form makes differentiation and integration far easier. But what about $kx - \omega t$? k is called the wave number and is defined as

$$k = \frac{2\pi}{\lambda}$$

and thus tells you how the phase changes for a given distance. for example if $\lambda = 2$

$$k = \frac{2\pi}{2} = \pi \text{ rad/m}$$

so kx tells how the phase changes as you move through space. in a similar fashion

$$w = \frac{2\pi}{T} = 2\pi f$$

and shows how the phase changes for a given time. Hopefully no example is needed this time. so wt tells you how the phase changes with time. But why do you subtract? Well first lets define a wave ψ such that

$$\psi(x) = A\cos(kx)$$

now if we translate it by a distance d

$$\psi(x) = A\cos(k(x - d))$$

and since $d = vt$

$$\psi(x) = A\cos(k(x - vt)) = A\cos(kx - kvt)$$

so

$$\begin{aligned}\psi(x) &= A\cos(kx - (\frac{2\pi}{\lambda})(\lambda f)t) \\ &= A\cos(kx - 2\pi ft) = A\cos(kx - wt)\end{aligned}$$

voila!

3 "Deriving" Schrödinger's equation

The Schrödinger equation cannot be formally derived from classical physics because it is a foundational postulate or Axiom of quantum mechanics. However we can show with circular logic (therefore not a derivation) how to arrive at the Schrödinger equation from the wave equation thus showing that the wave equation is a solution to the Schrödinger equation. First let us start by saying that

$$E = E_k + E_p = \frac{1}{2}mv^2 + u = \frac{p^2}{2m} + u$$

where u represents potential energy. Let us also define our wave ψ

$$\psi = e^{i(kx - wt)}$$

omitting the unnecessary constant A . now we can take the first and second spatial derivatives (derivatives with respect to x) using the chain rule

$$\begin{aligned}\frac{\partial \psi}{\partial x} &= (ik)e^{i(kx - wt)} = (ik)\psi \\ \frac{\partial^2 \psi}{\partial x^2} &= (ik)^2 e^{i(kx - wt)} = (ik)^2 \psi = -k^2 \psi\end{aligned}$$

and since $k = \frac{2\pi}{\lambda}$ we can rewrite k using DeBroglie's $p = \frac{h}{\lambda}$ as follows:

$$k = \frac{2\pi}{\frac{h}{p}} = \frac{p}{\hbar}$$

and thus

$$\begin{aligned}\frac{\partial^2 \psi}{\partial x^2} &= -\left(\frac{p^2}{\hbar^2}\right)\psi \\ -\hbar^2 \frac{\partial^2 \psi}{\partial x^2} &= p^2 \psi\end{aligned}$$

now if we multiply through our previously stated energy equation by ψ

$$E\psi = \frac{p^2 \psi}{2m} + u\psi$$

we can substitute for $p^2 \psi$ to get

$$E\psi = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + u\psi$$

giving us the time independent Schrödinger equation or TISE. To get the time dependent version we first take a temporal derivative

$$\frac{\partial \psi}{\partial t} = -i\omega e^{i(kx - \omega t)} = -i\omega \psi$$

we also know that $E = hf = \omega \hbar$ and by manipulating this and multiplying by $\frac{-i}{\hbar}$ we get

$$\frac{-i}{\hbar} E\psi = -i\omega t = \frac{\partial \psi}{\partial t}$$

so

$$E\psi = -i\omega t = \frac{\hbar}{-i} \frac{\partial \psi}{\partial t} = i\hbar \frac{\partial \psi}{\partial t}$$

which can now be substituted into our time independent equation to get

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + u\psi$$

The time dependent Schrödinger equation or TDSE.

4 Remarks

The Schrödinger equation is the fundamental equation of quantum mechanics similar to $F = ma$ in classical mechanics. It is used to describe how the wave-function ψ of a particle changes in space and time. I hope you have enjoyed.